Tidal potential measured on a coastal–inland optical-clock baseline

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Optical clocks measure gravitational-potential differences through their relative ticking rates.[1,2] Extending this measurement to changing potentials requires resolving small variations during the available observing intervals. Here we detect tidal changes in the potential difference between coastal Shanghai and inland Wuhan using a strontium clock and a ytterbium clock 692 km apart. We accumulated 545.9 hours of observations over a 353-day span, retaining an independent frequency level for each observing segment. The response to the combined solid-Earth and ocean-loading prediction is 0.96 ± 0.11 **(one statistical standard error). The changing lunar–solar phase also permits joint estimation of neighbouring twice-daily tidal groups with fixed internal amplitude ratios and phases. The lunar M_2+N_2 response is** 1.00 ± 0.13**, with phase and spectral structure consistent with the prediction. The neighbouring S_2+K_2 response is separately constrained within this model, but is more sensitive to disturbances at 12 hours. These observations show how astronomical phase allows tidal potential changes to be measured across interruptions in a distant optical-clock comparison.**

Earth's gravitational potential changes as the Moon and Sun move, the solid Earth deforms and ocean water is redistributed. These changes affect the rate at which a clock ticks relative to a clock elsewhere.[1,2] Comparing distant clocks therefore offers a way to observe changes in the potential difference between their locations. Laboratory redshift tests and remote optical-clock comparisons have established measurements of static potential differences.[3,4,5,6] The question addressed here is whether the same observable can reveal tidal changes over hours on a fixed, distant baseline.

Optical clocks connected by fibre networks can compare their rates over hundreds of kilometres.[7,8] Their sensitivity to gravity has two consequences. Changing gravitational shifts must be accounted for in precise frequency comparisons and time references, while those same shifts carry information about the changing Earth.[9,10,11,12,13] A dynamic measurement requires sufficient precision during the signal's variation, as well as a way to combine observations when the clocks are not operating continuously. Different observing runs need not retain the same frequency reference level.

Tides provide a signal whose timing remains calculable through such interruptions. The principal lunar semidiurnal tide M_2 repeats every 12.42 hours, close to the 12-hour principal solar tide S_2. A corresponding lunar maximum arrives about 50 minutes later on successive solar days, so the two waveforms change their relative phase. Even when a run's reference level is unknown, its measured changes can still be compared with the astronomical phase at each observation time. We use this information to detect tidal potential variation on the Shanghai–Wuhan baseline and then estimate the responses of neighbouring semidiurnal groups.

A baseline sensitive to tidal potential

We compared a strontium optical clock in coastal Shanghai with a ytterbium optical clock in inland Wuhan. The stations are 692 km apart and connected through Hefei by a 1,350-km phase-stabilized fibre path (Fig. 1a).[14] The reported clock-comparison instability follows the scale $5.4 \times 10^{-16}/\sqrt{\tau/(1\,\mathrm{s})}$, where τ is averaging time. This describes how frequency noise decreases with averaging in the reported comparison. Clock evaluations and field-fibre experiments establish the apparatus used for the

present measurement.[15,16,17]

Shanghai's proximity to the East China Sea gives the baseline a substantial sensitivity to ocean mass redistribution. Moving seawater changes both the gravitational attraction and the deformation of the solid Earth, including the position of a clock. The coastal and inland stations respond differently, adding an ocean-loading contribution to their potential difference. Our prediction combines this contribution with the direct astronomical tide and the solid-Earth response, using an IERS-based solid model and FES2022 ocean elevations (Methods).[18,19,20,21]

For M_2, the predicted differential fractional-rate amplitudes are 3.43×10^{-18} from the solid response and 1.42×10^{-18} from loading. The contributions differ in phase by 19.8°, giving a combined amplitude of 4.78×10^{-18}, or about $0.43\,\mathrm{m^2/s^2}$ in potential difference (Fig. 1b). Loading is about 30% of this combined M_2 amplitude. Its root-mean-square variation is 33.1% of the combined prediction on the weighted, segment-centred observation times. These quantities describe the predicted signal; the clock comparison tests the combined potential variation.

The September 2025–August 2026 record contains 3,299 accepted ten-minute means in 54 segments and eight observing periods. It represents 545.9 nominal member-hours on 43 UTC dates in eight calendar months, spread over 353 days (Fig. 1c). Summer contributes 273.9 hours; earlier observations add different lunar–solar phases (Fig. 1d). Sixteen summer segments overlap with the absolute frequency-ratio study of Hou et al.[14] The present analysis uses the extended record to estimate temporal tidal responses, without updating that absolute ratio.

Tidal variation in the clock record

The measured frequency ratio shows recurring variation at the predicted tidal phase (Fig. 2). The full record gives a response $A_T = 0.96 \pm 0.11$ to the complete predicted waveform (Fig. 3a). Here a single coefficient multiplies the combined solid-Earth and ocean-loading prediction, including its minor constituents, and unity denotes the predicted amplitude. This estimate uses all 3,299 ten-minute observations, rather than

the display averages or the selected time interval. The quoted uncertainty is one statistical standard error, using the larger of scaled least-squares and UTC-day-deletion errors. The measured response is consistent with the predicted tidal variation; uncertainty in the physical template is distinct from this statistical error.

A positive fractional-rate change means that the Wuhan Yb clock runs faster relative to the Shanghai Sr clock. In the potential convention used here, $y = -[\delta W_{\mathrm{Wuhan}} - \delta W_{\mathrm{Shanghai}}]/c^2$; the two axes in Fig. 2 express the same variation as a potential difference and a fractional rate.

Each segment has its own constant frequency level. We remove the weighted mean separately from the observations and from every prediction column, retaining their changes within the segment. The recorded UTC times then place those changes at the correct astronomical phases. Common physical response coefficients are estimated from all segments together, without joining their reference levels or filling the gaps. The algebra and the equivalent model with segment intercepts are given in Methods.

A view on the real time axis is provided by the 42.5-hour segment with the largest accepted member count (Fig. 2b,c). This selection depends on observing support, not agreement with the prediction. Three-hour display means make the variation visible, with identical averaging applied to the measurements and the unit prediction. The underlying 255 ten-minute means are also shown. For the full-record phase display, observations and the lunar template undergo the same removal of segment levels and other physical groups before averaging (Fig. 2a).

Amplitude, phase and nearby tidal groups

A single response tests the complete waveform but does not distinguish its astronomical groups. We therefore jointly estimate the lunar semidiurnal group M_2+N_2, the neighbouring group S_2+K_2 and an auxiliary group of approximately daily tides. Relative amplitudes and phases within each group remain fixed by the prediction. M_2 and N_2 arise from lunar forcing; S_2 is the principal solar semidiurnal tide, whereas

K_2 includes both lunar and solar forcing.

The grouped responses are $A_L = 1.00 \pm 0.13$ and $A_S = 0.93 \pm 0.36$, with a scaled least-squares correlation of -0.30 (Fig. 3a; Extended Data Fig. 1). The lunar group provides the more precise component estimate. The auxiliary daily group carries its correlations with both semidiurnal groups through the joint fit. All three coefficients are estimated from the same observations, after removing the fixed minor-constituent remainder once. The complete-waveform and grouped models thus answer different questions about one record.

The lunar phase can also be tested by adding a template shifted by a quarter-cycle. The fitted in-phase and quarter-cycle coefficients are 1.00 ± 0.13 and -0.10 ± 0.20, respectively (Fig. 3b). The latter, often called the quadrature response, measures departure from the predicted phase. Its agreement with zero supports the astronomical timing of the observed lunar variation.

The measured semidiurnal spectrum has local maxima at 22.361 and 23.147 µHz, near the prescribed M_2 and S_2 frequencies (Fig. 3c). The known astronomical frequency spacing is about 0.78 µHz; the reported scan maxima are 0.786 µHz apart. Because observations are interrupted, a signal at one frequency also produces secondary spectral structure through the sampling window. Existing single-line injections through that window give about 8% cross-power at the other principal frequency. Conditional scans remove the competing principal sinusoid before examining the target neighbourhood (Extended Data Fig. 2). These descriptions of frequency, phase and amplitude use the same record and do not represent independent repetitions of the detection.

Support across observing periods

Separate estimates for each of the eight periods show how observing duration and phase coverage affect the component constraints (Extended Data Fig. 3). Shorter records have broad intervals. The full-record response comes from the common physical fit, rather than a simple average of these separate estimates. Comparisons with

alternative reference-level choices are retained in Extended Data Fig. 4. The primary segment convention accommodates the absence of calibrated level transfers across the relevant interruptions.

The summer fibre round-trip monitor provides a simultaneous experimental control over 246.6 member-hours. On identical clock and monitor support, adding this measurement changes the lunar coefficient by 0.0061 ± 0.0063 and the neighbouring coefficient by 0.00594 ± 0.01453 (Extended Data Fig. 5). These small paired changes constrain the monitored transmission path during the overlap. The tidal estimates themselves use the full clock record.

The lunar and neighbouring responses differ in their sensitivity to possible daily disturbances. A shared 24-hour sine–cosine pair gives a lunar coefficient of 1.04 ± 0.13 and a complete-template coefficient of 1.03 ± 0.10. Giving each period a freely phased 12-hour pair yields $A_T = 0.72 \pm 0.22$ in the complete-template model. In the grouped model it gives $A_L = 0.75 \pm 0.24$ and $A_S = -5.10 \pm 15.01$. These are sensitivity designs, not measurements of laboratory disturbances. They show that the neighbouring group's physical attribution depends much more strongly on the treatment of variation at the same frequency. The observed lunar response remains the clearest component of the tidal detection; full definitions and intervals are retained in Methods.

Discussion

The Shanghai–Wuhan frequency comparison detects changes in gravitational potential difference through changes in the relative rates of two distant clocks. The combined-waveform response quantifies this measurement, while the lunar amplitude, phase and spectral structure identify its clearest astronomical component. Tidal variation is therefore accessible as an observable in this comparison, alongside the static potential difference that remote clocks can measure. The experiment connects a frequency measurement between fixed laboratories to the changing gravitational environment across their baseline.

The known astronomical phase is what allows interrupted observations to contribute to this temporal measurement. Each segment can retain an unknown constant while its internal variation contributes to a common physical response. Observations on different dates add different combinations of lunar and solar phase, allowing neighbouring semidiurnal groups to be constrained jointly. This is useful for clock comparisons whose operation is intermittent and whose reference levels are not transferred across every interruption. The demonstrated accumulation relies on predictable phase; an unanticipated transient would instead require sufficient precision during the event itself.

The coastal–inland geometry makes ocean loading a substantial part of the potential variation being tested. The measured response applies to the combined potential, and the loading contribution is not independently separated from the solid response. Observations that more strongly distinguish their patterns could turn this sensitivity into a separate loading test. More continuous operation and independent instrumental records near 12 hours would also strengthen attribution of the neighbouring semidiurnal response. These are specific extensions of a measurement that already recovers tidal potential variation from interrupted clock observations. Related tidal-potential results have been reported independently by a European optical-clock network in a contemporaneous arXiv preprint.[22]

References

1. Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E. & Schmidt, P. O. Optical atomic clocks. *Rev. Mod. Phys.* **87**, 637–701 (2015). https://doi.org/10.1103/revmodphys.87.637
2. Mehlstäubler, T. E., Grosche, G., Lisdat, C., Schmidt, P. O. & Denker, H. Atomic clocks for geodesy. *Rep. Prog. Phys.* **81**, 064401 (2018). https://doi.org/10.1088/1361-6633/aab409
3. Chou, C. W., Hume, D. B., Rosenband, T. & Wineland, D. J. Optical Clocks and Relativity. *Science* **329**, 1630–1633 (2010). https://doi.org/10.1126/science.1192720
4. McGrew, W. F. et al. Atomic clock performance enabling geodesy below the centimetre level. *Nature* **564**, 87–90 (2018). https://doi.org/10.1038/s41586-018-0738-2
5. Grotti, J. et al. Geodesy and metrology with a transportable optical clock. *Nat. Phys.* **14**, 437–441 (2018). https://doi.org/10.1038/s41567-017-0042-3
6. Takano, T. et al. Geopotential measurements with synchronously linked optical lattice clocks. *Nat. Photon.* **10**, 662–666 (2016). https://doi.org/10.1038/nphoton.2016.159
7. Lisdat, C. et al. A clock network for geodesy and fundamental science. *Nat. Commun.* **7**, 12443 (2016). https://doi.org/10.1038/ncomms12443
8. Riehle, F. Optical clock networks. *Nat. Photon.* **11**, 25–31 (2017). https://doi.org/10.1038/nphoton.2016.235
9. Voigt, C., Denker, H. & Timmen, L. Time-variable gravity potential components for optical clock comparisons and the definition of international time scales. *Metrologia* **53**, 1365–1383 (2016). https://doi.org/10.1088/0026-1394/53/6/1365
10. Denker, H. et al. Geodetic methods to determine the relativistic redshift at the level of 10^{-18} in the context of international timescales: a review and practical results. *J. Geod.* **92**, 487–516 (2018). https://doi.org/10.1007/s00190-017-1075-1
11. Schröder, S., Stellmer, S. & Kusche, J. Potential and scientific requirements of optical clock networks for validating satellite-derived time-variable gravity data. *Geophys. J. Int.* **226**, 764–779 (2021). https://academic.oup.com/gji/article/226/2/764/6212226
12. Dimarcq, N. et al. Roadmap towards the redefinition of the second. *Metrologia* **61**, 012001 (2024). https://doi.org/10.1088/1681-7575/ad17d2
13. Lindvall, T. et al. Coordinated international comparisons between optical clocks connected via fiber and satellite links. *Optica* **12**, 843–852 (2025). https://doi.org/10.1364/OPTICA.561754
14. Hou, L. et al. Yb/Sr optical clock comparison at 2.2×10^{-18} uncertainty over a 1350-km fiber link. *arXiv*, 2610.34250v1 (2026). https://arxiv.org/abs/2610.34250v1
15. Jia, Z.-P. et al. Improved systematic evaluation of a strontium optical clock with uncertainty below 1×10^{-18}. *Metrologia* **63**, 025002 (2026). https://doi.org/10.1088/1681-7575/ae449e
16. Zhu, Q. et al. Ytterbium lattice clock with systematic uncertainty of 1.3×10^{-18} and instability at the 10^{-19} level. *arXiv*, 2606.10514v2 (2026). https://arxiv.org/abs/2606.10514v2
17. Chen, F.-X. et al. 10^{-21}-level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light Sci. Appl.* **15**, 276 (2026). https://doi.org/10.1038/s41377-026-02299-1
18. Petit, G. & Luzum, B. (eds) IERS Conventions (2010). *IERS Technical Note* **36** (2010). https://iers-conventions.obspm.fr/content/tn36.pdf
19. CNES. FES2022 (Finite Element Solution) Tidal model (Version 2024). CNES https://doi.org/10.24400/527896/A01-2024.004 (2024).
20. Farrell, W. E. Deformation of the Earth by surface loads. *Rev. Geophys.* **10**, 761–797 (1972). https://doi.org/10.1029/RG010i003p00761
21. Han, D. & Wahr, J. The viscoelastic relaxation of a realistically stratified earth, and a further analysis of postglacial rebound. *Geophys. J. Int.* **120**, 287–311 (1995). https://doi.org/10.1111/j.1365-246x.1995.tb01819.x
22. Stahl, K. et al. Optical clock network tracks tidal variations of the gravity potential. *arXiv*, 2610.39635v1 (2026). https://arxiv.org/abs/2610.39635v1

Figure legends

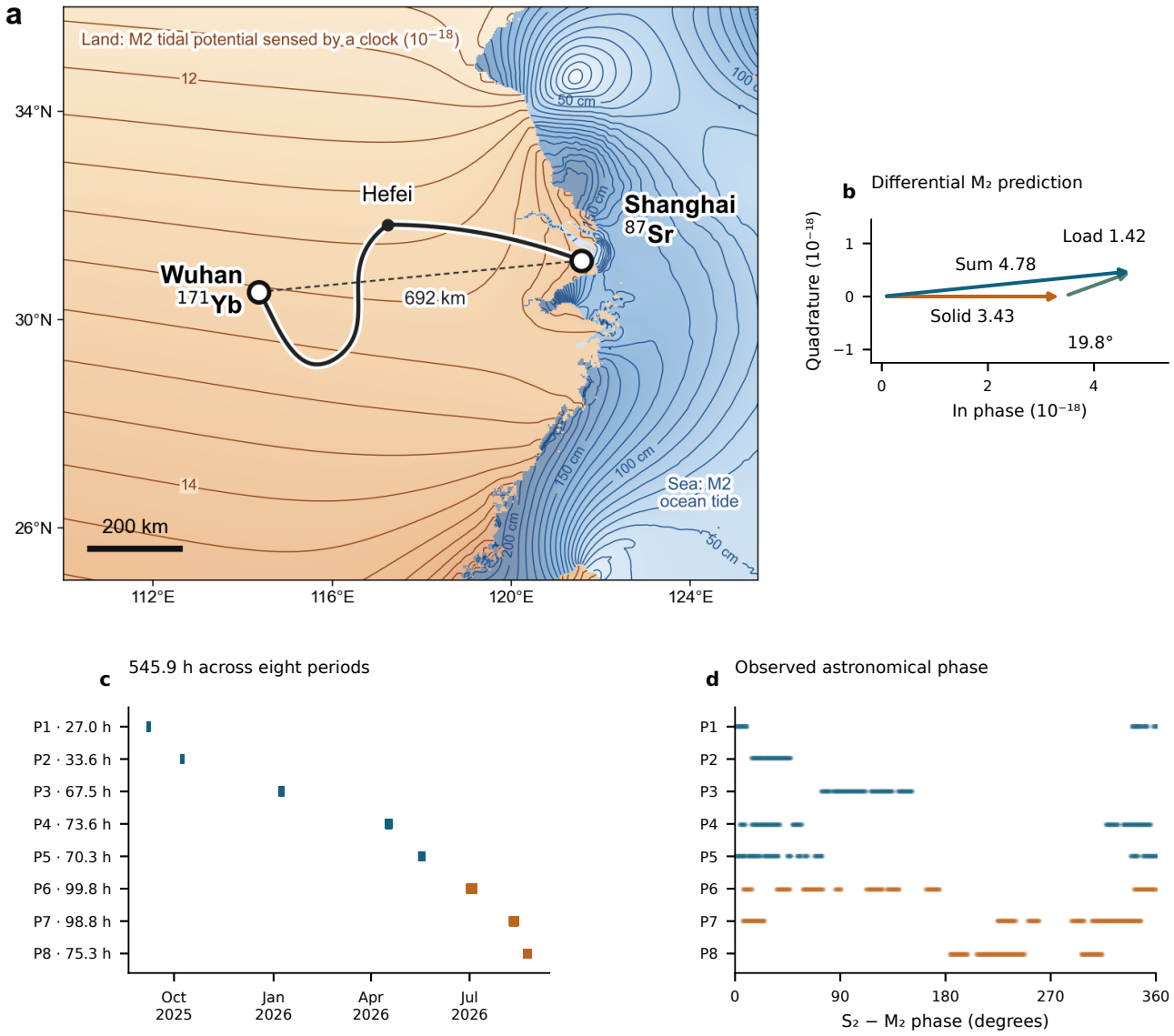


Fig. 1 | A coastal–inland baseline and its tidal prediction. **a,** Shanghai Sr and Wuhan Yb connected by the fibre route through Hefei. The dashed line marks the 692-km baseline. The supplied map illustrates an M_2 fractional clock-rate field over land in 10^{-18} and ocean-tide height contours at sea in centimetres. These are different physical quantities. **b,** Differential solid, loading and combined M_2 phasors from the frozen numerical prediction, with equal data-axis scales; the combined 4.78×10^{-18} corresponds to about $0.43\,\mathrm{m}^2/\mathrm{s}^2$. **c,** Actual UTC sampling and accepted member-hours in eight periods. **d,** S_2−M_2 phase at those same observations, including nodal corrections. P6–P8 are summer. The map fields and phasors are predictions; the lower panels describe the measured observation window.

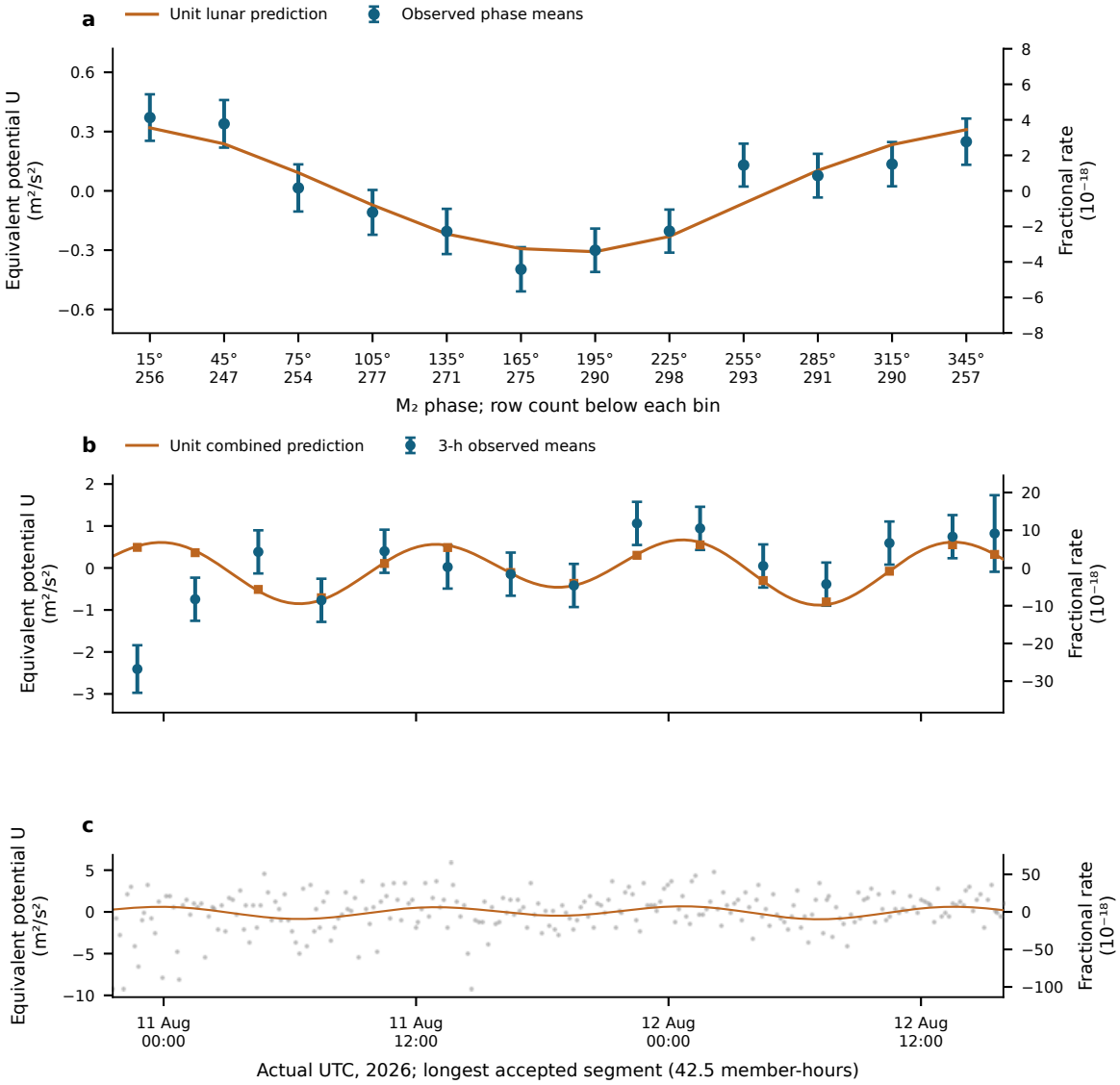


Fig. 2 | Tidal variation in the measured clock rates. **a,** Twelve M_2-phase means of all 3,299 ten-minute observations. The fixed minor remainder is removed, and data and unit lunar prediction undergo identical weighted projection against segment levels and the other physical groups. Error bars are one working standard error with nuisance propagation; means are correlated. Row counts appear beneath the phase labels. **b,** The segment with the largest accepted member count, shown at real UTC. Points are fixed-UTC three-hour means after independent weighted centring of data and prediction. Squares apply the same averaging to the unit combined prediction; the line evaluates it at the observed ten-minute times. Error bars propagate centring. **c,** Every original ten-minute mean in that segment, with its complete vertical range. No empirical daily sinusoid or fit-dependent offset is removed. Left axes use $U = c^2 y = -\Delta W$; right axes give fractional rate. Panels are views of the same measurements, and display averaging does not change the primary sample or inference.

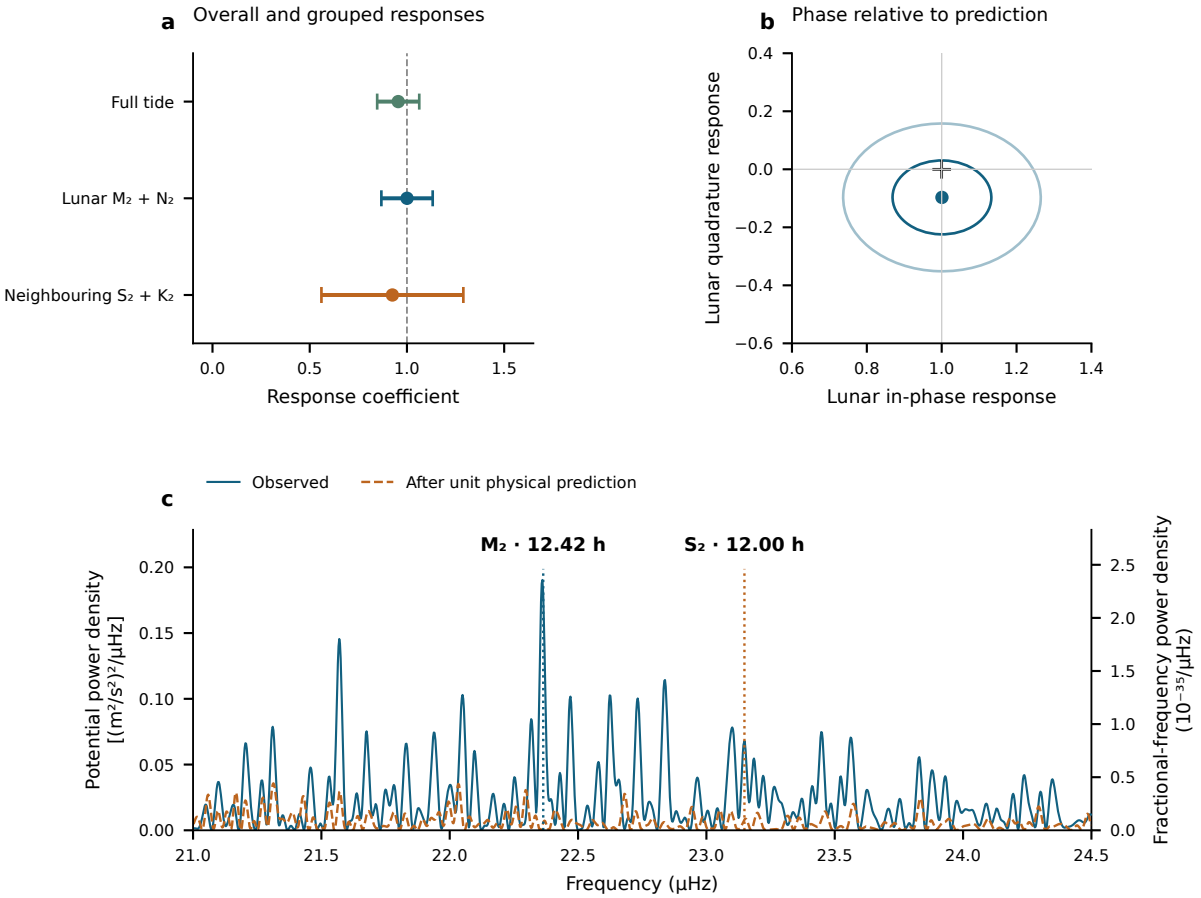


Fig. 3 | Amplitude, phase and frequency of the tidal response. a, The single complete-template response, followed by the jointly fitted lunar and neighbouring semidiurnal coefficients. Error bars are one statistical standard error, the larger of scaled LS and UTC-day-deletion jackknife errors. The estimates share the same observations and weights. **b,** Lunar in-phase and quadrature fit; the cross marks $(1, 0)$. Contours use scaled-LS covariance radii one and two. The marginal working errors reported in the text are 0.13 and 0.20; the scaled-LS quadrature error is 0.127. These are not independently calibrated joint confidence levels. **c,** The measured 21–24.5 μHz spectrum and the spectrum after subtracting the fixed unit physical prediction. The scan retains all 3,299 original UTC rows and gaps, with equal weights and 54 segment constants. Vertical markers are the prescribed M_2 and S_2 frequencies. Left and right density axes obey $S_W = c^4 S_y$. Peaks, phase and amplitudes use the same record and do not add independent detection significance. Sampling cross-power and conditional scans are given in Extended Data Fig. 2; joint group regions appear in Extended Data Fig. 1. The full-template and three-group coefficients come from two fits to the same observations; they are not averaged.

Methods

Observations, frequency sign and averaging

The Shanghai and Wuhan coordinates used by the frozen prediction are 31.130° N, 121.570° E and 30.529° N, 114.357° E. The measured ratio is Wuhan Yb/Shanghai Sr. With positive geodetic potential W, its tidal fractional variation is $-[\delta W_{\mathrm{Wuhan}} - \delta W_{\mathrm{Shanghai}}]/c^2$. The period-relative beat conversion is $-d_p(f_{\mathrm{beat}} - f_{\mathrm{ref},p})/F_{1550}$, where $F_{1550} = 1.933992 \times 10^{14}$ Hz and $d_p = 2$ in 2025 and 1 in 2026. Thus a 2026 beat deviation of 1 Hz has fractional magnitude 5.17×10^{-15}. The 2025 division factor is applied before forming the small clock-rate variation.

The supplied valid-interval table defines 54 segments in 29 operating groups and eight periods. Counter members are read across midnight before binning. Beijing-time labels are converted to UTC. Summer group corrections follow the decimal-arithmetic frequency-chain calculation; early absolute correction entries remain incomplete. Their constant part is represented by segment intercepts, while any time variation within a segment remains a possible systematic contribution. Fitted intercepts are not used to join the runs for long-term stability estimation.

The membership record contains 1,966,908 nominal one-second rows before the minimum-bin-support cut. Three boundary rows reconcile its difference from the printed interval total. Means are formed in nominal 600-s bins separately within each segment and require at least 180 retained members. The 3,299 accepted means contain 1,965,327 members, or 545.924 h. There are 3,291 unique bin centres: eight centres contain separate partial means from different segments, with disjoint members. The 353.39-day span contains observations on 43 UTC dates. The inherited valid mask is retained; this work neither applies an additional triangular filter nor selects points using tidal residuals. Exact raw-file memberships and hashes are retained in the controlled archive.

Hou et al. reported a combined-template tidal response of 0.87 ± 0.15 from the 16 overlapping summer segments.[14] That estimate belongs to the earlier data selection and model and is not an independent repetition of the present result.

What the constituent names mean

Semidiurnal tides produce approximately two cycles per day. M_2 is the principal lunar constituent; N_2 describes an additional lunar variation associated with orbital ellipticity. S_2 is the principal solar semidiurnal constituent, whereas K_2 contains lunar and solar forcing. The names identify astronomical frequencies, not separate instruments or independent experiments. The diurnal group contains K_1 (lunisolar), O_1 (principal lunar), P_1 (principal solar) and Q_1 (lunar elliptic). Their periods are listed in Extended Data Table 1.[23] Internal relative amplitudes, phases and nodal modulation are fixed within each physical group. A group response therefore tests that complete specified pattern.

Combined solid and loading potential

The physical target includes the direct astronomical potential, the Earth's induced potential and the effect of clock displacement. The selected frozen solid sequence implements these through the IERS radial displacement u and a fixed effective-potential conversion,

$$y_{\mathrm{solid}} = -\frac{1 + k_2 - h_2}{h_2}\frac{g(u_{\mathrm{Wuhan}} - u_{\mathrm{Shanghai}})}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\,\mathrm{m/s^2}$. This is a displacement-based approximation to the combined solid potential. The same conversion also acts on degree-three and frequency-dependent displacement contributions in the supplied sequence. A previous implementation comparison found an approximately 18% K_1 amplitude deficit relative to a potential-weighted reference. The daily response is accordingly not used to infer a Love number. The quoted statistical errors are conditional on this solid template. A same-sampling potential-weighted comparison has not been completed for these estimates, so the K_1 discrepancy is not assigned as an error on the semidiurnal response.

For loading, FES2022 sea-surface elevations ζ are convolved with the effective-

potential kernel

$$\delta W_{\mathrm{load}} = \int \rho_w \zeta(\mathbf{r}', t) G_W(\psi)\, \mathrm{d}A, \qquad G_W(\psi) = \frac{ga}{M_E} \sum_n (1 + k'_n - h'_n) P_n(\cos\psi).$$

The three terms represent direct water attraction, induced potential and clock displacement. PREM load Love numbers are retained through degree 696 with the archived high-degree continuation. Eighteen loading constituents are synthesized with their astronomical nodal corrections. The complex convention is $f\,\mathrm{Re}[C\exp(-i\phi)]$; reversing the exponential sign requires conjugating C. The present analysis reads the frozen constituent and time-series tables rather than substituting an older loading sequence.

The eight named constituents form the three physical groups. Minor terms remain once as a fixed remainder r_0. On the actual weighted segment-centred sampling, solid, loading and total root-mean-square variations are respectively 2.928, 1.328 and 4.011×10^{-18}. Their ratio 0.331 is an amplitude-norm ratio, not an additive fraction of total power. A separate loading coefficient is not estimated here; the measured group responses test the combined prediction.

Segment reduction and the two estimands

Let W_s be the original diagonal weight matrix in segment s, and define $\langle v \rangle_s = (\mathbf{1}^T W_s v)/(\mathbf{1}^T W_s \mathbf{1})$. For each observation vector and each prediction column we form $\tilde{v}_s = v_s - \mathbf{1}\langle v \rangle_s$. This operation uses sample times and weights but no fitted tidal amplitude. For a physical design matrix T_s, the common coefficients solve

$$\widehat{A} = \left(\sum_s \widetilde{T}_s^T W_s \widetilde{T}_s \right)^{-1} \sum_s \widetilde{T}_s^T W_s \tilde{y}_s.$$

The segment matrices and vectors contain the information needed for this estimate. All 54 segments enter the same fit; separately fitted segment amplitudes are not

averaged. An independent full-design QR fit with 54 intercepts reproduces the three coefficients to within 3×10^{-15}. Recomputing the segment means after every deletion also reproduces the original standard errors. No segment level is transferred to create a continuous, calibrated long-term record.

For the full-response estimate, y is the observed fractional rate before removal of the minor remainder. We fit

$$y_i = c_{s(i)} + A_T T_i + \epsilon_i, \qquad T_i = s_{L,i} + s_{S,i} + s_{D,i} + r_{0,i}.$$

The single coefficient multiplies the complete prediction. The restricted comparison retains only the 54 segment constants. For component estimation, we subtract the fixed minor remainder once and fit

$$y_i - r_{0,i} = c_{s(i)} + A_L s_{L,i} + A_S s_{S,i} + A_D s_{D,i} + \epsilon_i.$$

The full-response model tests the combined waveform, whereas the grouped model allows the three astronomical bands to respond differently. Both use $W_{ii} = 1/\sigma_i^2$ with the period-dependent Allan-based uncertainties defined below.

For Extended Data Fig. 3, we repeat the same three-group estimation separately on each of P1–P8. The data, fixed theoretical remainder and Allan-based weights are inherited from the frozen primary design; each period retains its original segment constants. The UTC-day jackknife is recomputed using only the dates present in that period. All period fits and day-deletion replicates retain full column rank and were independently checked against an explicit segment-intercept QR calculation. The individual-period errors use the same working standard-error rule as the full record.

The daily group is an auxiliary physical component; its coefficient remains 0.58 ± 0.54 (Extended Data Table 2). The interval and correction records do not establish calibrated zero transfer at the 46 boundaries where an eight-period intercept would be shared. They document valid observing times but not signed boundary transfers

with uncertainties. The segment convention follows that experimental information.

Construction of bin uncertainties

For ten-minute bin i, the assigned uncertainty is

$$\sigma_i = \frac{\alpha_{p(i)}}{\sqrt{n_i}},$$

where n_i counts retained nominal one-second members and α_p is the one-second fractional-frequency noise level of period p. In this convention the corresponding white-frequency instability is $\alpha_p/\sqrt{\tau/(1\,\mathrm{s})}$. The machine-readable source data provide all eight period coefficients, bin member counts and reconstructed uncertainties.

The noise calculation uses original segments containing at least 8,192 retained members. The fixed unit tidal prediction is evaluated at member timestamps and subtracted after centring within each segment. Overlapping Allan deviations are calculated at 256, 512, 1,024, 2,048 and 4,096 nominal seconds, using the acquisition's one-second member convention. At each averaging time the deviations are averaged across eligible segments, weighted by their member counts. The geometric mean of these averaged deviations multiplied by $\sqrt{\tau/(1\,\mathrm{s})}$ gives α_p. No Allan window crosses a segment boundary. These coefficients determine relative fitting weights; residual scaling and day deletion determine the reported statistical errors.

The instability benchmark quoted with Fig. 1a was reported for the spliced summer comparison.[14] It is not used to assign the present weights, which are reconstructed from the period coefficients and retained member counts. No stability estimate from joined, fit-adjusted levels is used in this analysis.

Statistical errors and conditional coverage

For each group the reported standard error is the larger of residual-scaled least squares and leave-one-UTC-day-out jackknife errors. Deletions remove a calendar day, refit all remaining coefficients and drop empty intercept columns. The jackknife

variance is $(G-1)G^{-1}\sum_g(\hat{A}_{(-g)}-\overline{A})^2$. Period-deletion errors are recorded separately. Extended Data Table 2 gives all components and the same-data intercept comparisons. Observations are repeated measurements of one clock pair, not thousands of independent experiments.

Conditional simulations apply both designs to the same 3,299-row sampling and heteroscedastic uncertainties. Each of four scenarios has 1,000 realizations: independent Gaussian noise; equal variance contributions from Gaussian white noise and a 6-h Ornstein–Uhlenbeck process within segments; the corresponding 24-h process within periods; and standardized Student-t_5 noise. The overall scale is the observed segment-model residual scale; all true physical response coefficients are one. Each realization recomputes both components of the standard-error rule. Seed 2026110617 fixes the simulation. Coverage of $\hat{A}\pm 1.96u$ is 95.1–97.4% for the segment-model lunar coefficient and 95.4–96.8% for its neighbouring coefficient; Monte Carlo standard errors are about 0.4–0.7 percentage points (Extended Data Table 3). These are tests under declared noise laws, not a calibration against every possible disturbance.

The full-template coefficient uses the same standard-error rule. Its scaled-LS, day-deletion, day-cluster CR1 and period-deletion standard errors are respectively 0.0973, 0.1082, 0.1033 and 0.1216. The CR1 sandwich uses 43 UTC-day clusters and the finite-sample factor $[G/(G-1)][(n-1)/(n-K)]$, with $K=55$. The working interval $\widehat{A}_T\pm 1.96u$ is 0.743–1.167. This is an explicitly labelled working interval, rather than a confidence statement valid for every possible noise process.

For targeted zero-response tests, we compare the weighted RSS of nested models on the unchanged rows. The full-template test adds one entire prediction to segment constants. The neighbouring-group test adds one S_2+K_2 coefficient while retaining the lunar and daily groups and the constants. Noise is generated on the true window with the original heteroscedastic uncertainties and the observed full-model residual scale. Each of the four declared noise scenarios has 20,000 realizations; seeds 2026110711 and 2026110709 fix the full-template and neighbouring-group simula-

tions. The exact residualized-template projection computes the RSS improvement, and direct restricted/full refits check that calculation. The full-template statistic is 110.353; none of the 20,000 draws in any scenario reaches it. The plus-one tail estimate is $1/20,001$ in each scenario, not zero. For the neighbouring statistic 10.943, the corresponding tail estimates are 0.00185, 0.12559, 0.01435 and 0.00205 in the listed noise-scenario order.

Restricted period-wise residual sign flips enumerate all $2^8 = 256$ patterns, preserving each period's internal structure. The diagnostic tail fractions are 2/256 for the full response and 16/256 for the added neighbouring group. They rely on period independence and symmetry and are not exact experimental randomization tests. The two-sided resolution is 2/256; neither enumeration is converted to a Gaussian sigma value.

The noise scenarios are conditional stress tests, not measured noise laws. In separate 2,000-draw diagnostics per law, the observed within-segment lag-one residual correlation is −0.148. The simulated 95% ranges are −0.054–0.015 for Gaussian noise, 0.216–0.347 for the 6-h mixture, 0.072–0.202 for the 24-h mixture and −0.051–0.017 for t_5. Thus the positively correlated mixtures do not describe the observed short-lag structure; neither do the other two laws fully reproduce it. No scenario is selected because of its tidal tail probability. No simulated record is added to the observations.

Extended Data Fig. 1 gives conditional two-group regions: the first 10,000 draws estimate error covariance and the empirical 95% Mahalanobis threshold; the other 10,000 assess coverage. Held-out joint coverage is 94.3–94.8%, with binomial Monte Carlo standard errors of about 0.22 percentage points and additional calibration uncertainty. These nominal regions describe the specified laws and do not replace the reported marginal working errors with a universally calibrated joint confidence region.

The phase-relaxed model adds a lunar component shifted by a quarter-cycle while retaining the prescribed M_2/N_2 relationship. Covariance contours use scaled least squares. Subset and deletion fits refit the same model on their retained rows. Shared-

data estimates are not averaged. The simulation coverage refers to two-sided intervals; the reported ± values are standard errors.

Phase means, spectral density and sampling sidelobes

For the phase plot, N contains segment constants and the two non-lunar groups. We project both $y - r_0$ and the unit lunar prediction with $M_N = I - N(N^T W N)^{-1} N^T W$, and average in twelve fixed M_2 phase bins. If B is weighted bin averaging, the mean covariance is $B M_N C_y M_N^T B^T$, where C_y is the scaled working observation covariance. It retains correlations between means and propagates nuisance estimation. The response is estimated from the original ten-minute means, before phase binning.

For the actual-time display, the representative segment maximizes accepted nominal member count, with segment ID breaking ties. Segment 51 contains 255 ten-minute means and 153,000 members. Its 42.5 member-hours extend from 10 August 2026 21:35 UTC to 12 August 15:55 UTC. Three-hour display bins follow fixed UTC boundaries and remain inside the segment. All 255 means are assigned; none is removed to improve the display. Weighted averaging is identical for observed and unit-predicted centred values. If a display bin has total weight w_b and its segment has total weight w_s, the centred-observation variance is $\hat{\sigma}^2(1/w_b - 1/w_s)$. This propagates the segment-mean operation and creates correlations between display means. The curve at ten-minute timestamps is a sampled physical prediction, not gap filling. The potential display uses $U = c^2 y$ and the right axis retains the fractional rate.

Figure 3c preserves the independently specified equal-weight, 54-intercept scan of the unchanged observed means, rather than replacing it with the amplitude fit's weights. Every trial sine–cosine pair and the data are projected against the same constants. For a reduction in squared residuals $\Delta\mathrm{RSS}(f)$ with dimensionless y, the nominal one-sided projection density is $S_y(f) = 600\,\mathrm{s}\,\Delta\mathrm{RSS}(f)$ per Hz, or 10^{-6} times that value per µHz. Potential density is $S_W = c^4 S_y$; amplitude density is $c^2\sqrt{S_y}$. Uniform-grid checks reproduce the conventional one-sided Fourier density. For interrupted data this normalization is a stated projection density, not a Parseval-calibrated variance over a filled year.[24]

The frequency step is 0.002729 µHz; the inverse calendar span is 0.032752 µHz. Fine plotting steps do not improve physical resolution. The listed peak positions are scan maxima; no constituent-frequency fit or frequency-confidence interval is reported. M_2-only and S_2-only astronomical injections use the same actual times, nodal factors and equal-weight segment projection. Power at the opposite principal frequency, divided by power at the injected frequency, gives 0.08027 and 0.08149. Conditional scans project out a freely phased sinusoid at the other frequency before scanning the target neighbourhood. Amplitude interference means that such corrections act on sampled waveforms or complex coefficients, not by subtracting scalar power spectra. Gaps alone do not predict the realization of measurement noise. Residual density can describe remaining measurement fluctuations, but includes possible model mismatch and cannot be called a pure instrumental noise floor.

Instrument and daily-response checks

The matched monitor calculation uses 1,482 rows with 887,882 nominal members, or 246.634 h, in P6–P8. Both baseline and augmented fits have identical members, segment labels and weights. Paired day-deletion replicates propagate uncertainty in their difference. Extended Data Fig. 5 reports the response changes and dates. This comparison concerns the monitored transmission route during the measured overlap.

On the selected data and model, a common 24-h sine–cosine pair gives $A_L = 1.036 \pm 0.133$; separate 24-h pairs in each period give 1.003 ± 0.136. These daily sensitivity designs address a periodicity motivated by temperature and operating schedules, without assigning an independently measured thermal amplitude. A shared 12-h pair gives lunar 0.994 ± 0.132 and neighbouring 2.931 ± 1.011. Separate pairs in each period give 0.751 ± 0.242 and -5.100 ± 15.008. The normalized design condition number grows from 2.37 to 6.90 and 122.54, respectively; all designs retain full numerical rank. These terms test same-frequency identifiability and are not independently measured disturbances. Extended Data Table 4 reports both groups and their 1.96-SE working intervals.

The full-template sensitivity fits add either one shared 24-h sine–cosine pair or a separate 12-h pair in each of the eight periods. The basis functions are evaluated at the recorded UTC bin centres. All 3,299 observations, 54 segment constants, original weights and the full minor remainder are retained. Each model recomputes scaled least-squares and 43-day-deletion errors; Extended Data Table 5 lists both components. These scalar-response fits are distinct from the grouped models in Extended Data Table 4.

AI-assisted preparation

OpenAI Codex assisted English drafting, code preparation and checking, numerical replay, and plotting from derived experimental tables using Python. The geographical map in Fig. 1a was supplied by the authors and retained unchanged. No observations were generated to replace or fill the experimental record. This manuscript revision reorganized the existing narrative and displays without rerunning fits or data validation. [Authors to confirm the tool and version details, their verification of the code, source data, statistics, figures and citations, and responsibility for the final work.]

Data availability

The author review package contains the 3,299 derived means, fixed templates, noise weights, figure source tables and immutable input hashes. The original counters and membership ledger are held in the controlled experimental archive. [Authors to supply an editor-accessible location and confirm the permanent data-access statement and repository identifier before submission.]

Code availability

The package contains the independent QR replay, same-data design comparison, conditional simulations, spectral calculations, covariance propagation and figure scripts, with software versions and a reproducible entry point. [Authors to confirm the code licence and permanent repository identifier before submission.]

Methods references

23. NOAA. Harmonic constituents. *NOAA Tides and Currents*. https://tidesandcurrents.noaa.gov/harcon.html?id=8577330 (accessed 6 November 2026).
24. VanderPlas, J. T. Understanding the Lomb–Scargle periodogram. *Astrophys. J. Suppl. Ser.* **236**, 16 (2018). https://doi.org/10.3847/1538-4365/aab766

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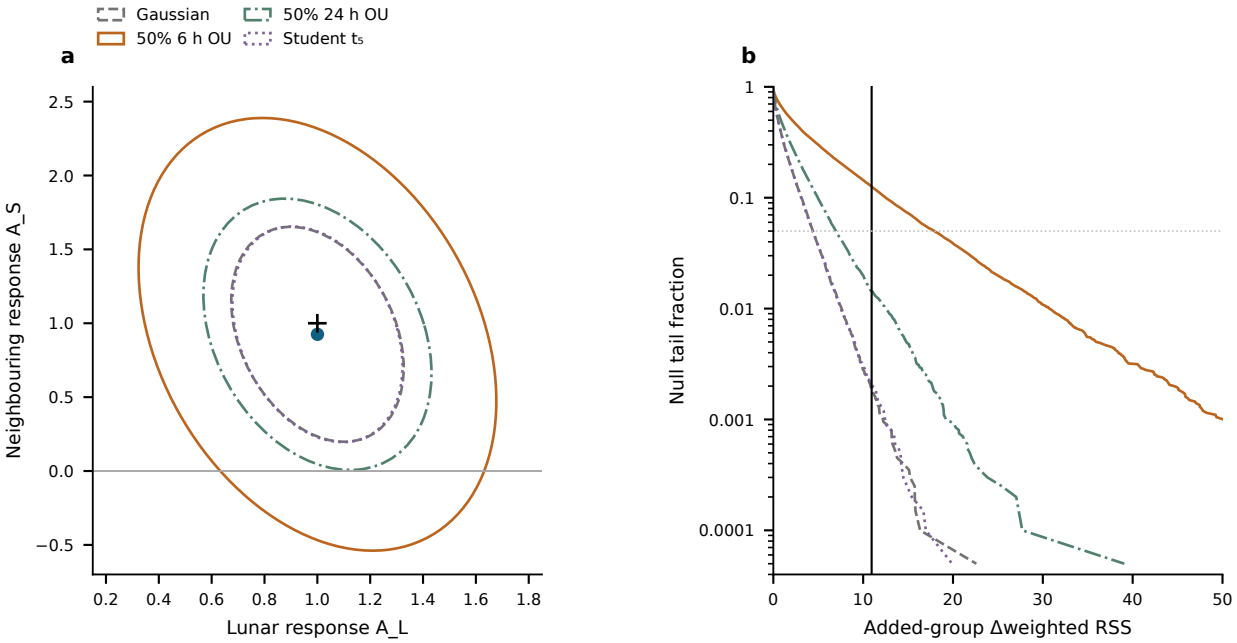
Competing interests

[Declaration to be supplied by the authors.]

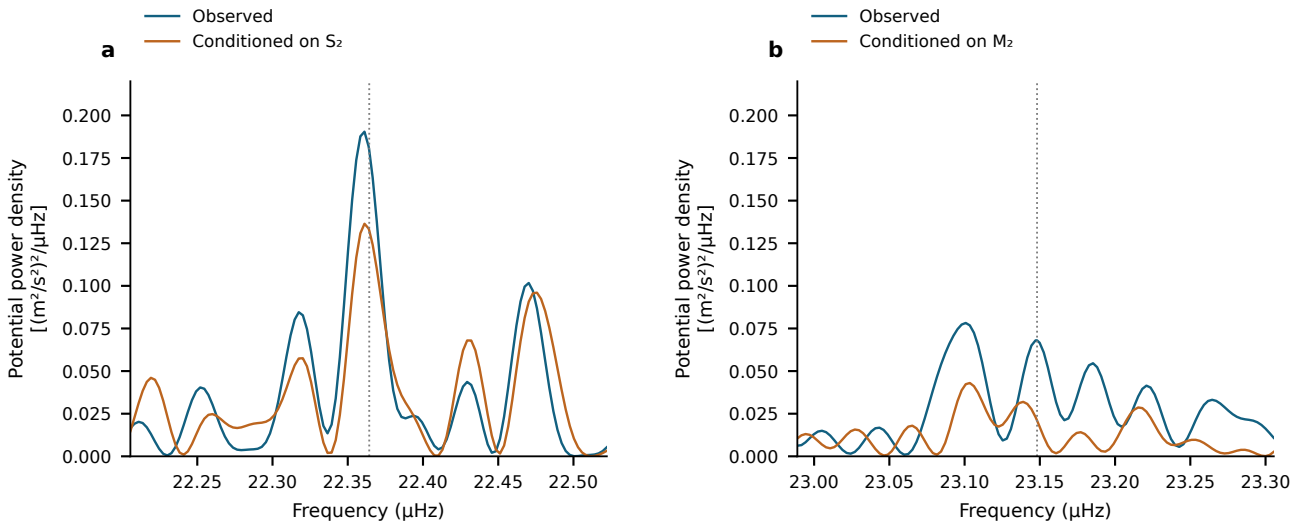
Additional information

Correspondence: [Corresponding author names and email addresses to be supplied; mark them with an asterisk in the author list.]

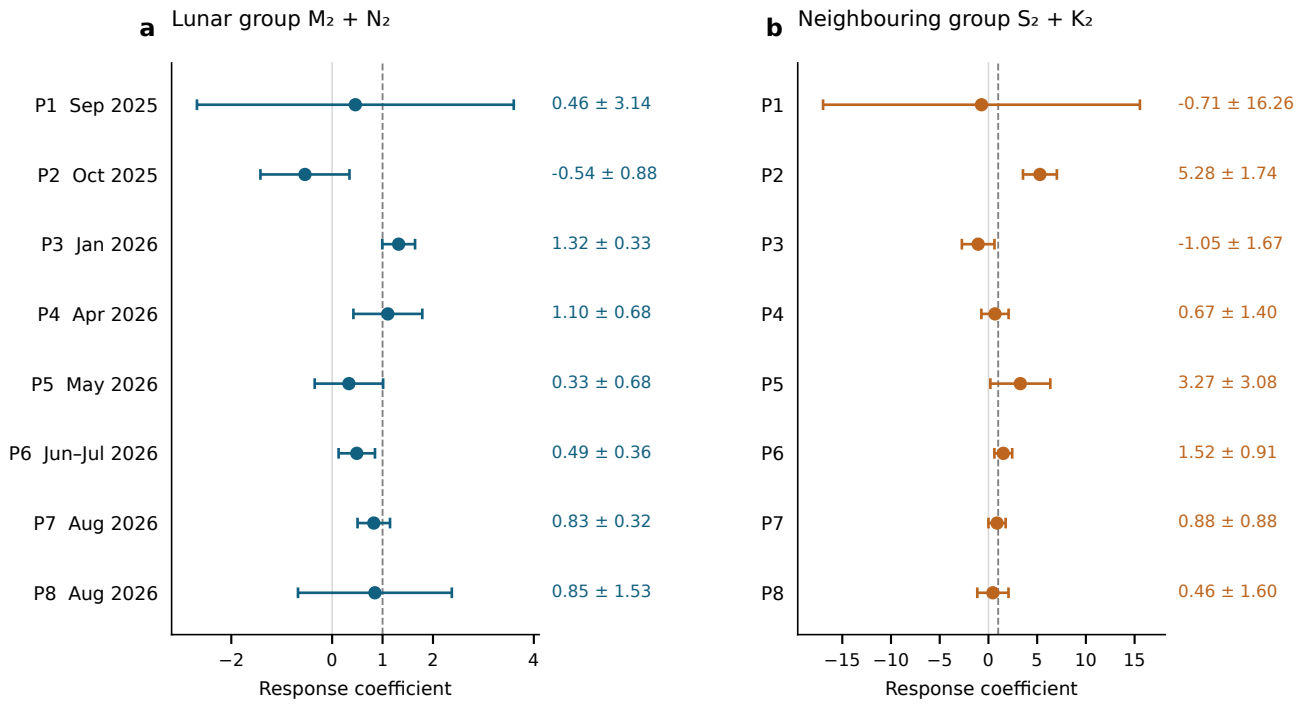
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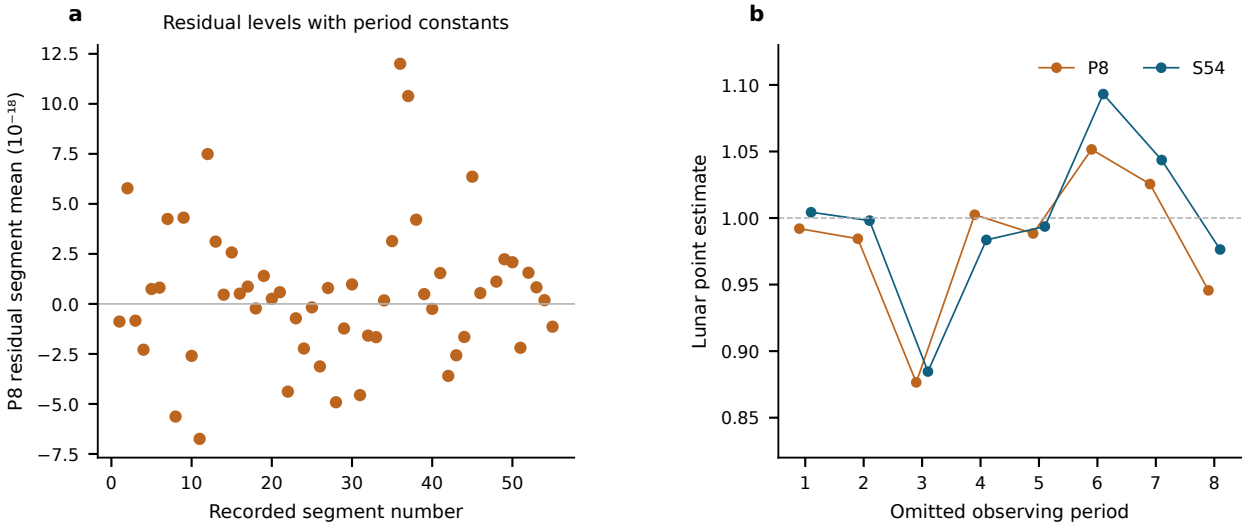
Extended Data Fig. 1 | Conditional uncertainty of the neighbouring response. a, Nominal 95% joint regions for (A_L, A_S) under four declared noise laws on the original 3,299-row design. The point is the observed estimate and the cross is $(1, 1)$. Each region uses 10,000 calibration and 10,000 validation simulations; held-out coverage is 94.3–94.8%. The regions are conditional on the specified laws, none of which fully matches the measured short-lag residual structure. **b,** Null tail curves for the added S_2+K_2 coefficient from 20,000 simulations per law. The vertical line marks the observed weighted-RSS improvement; the dotted horizontal line is 0.05. Simulations are separate from the measured data. The declared scenarios provide sensitivity information, not four additional experiments or an independently validated laboratory-noise model.



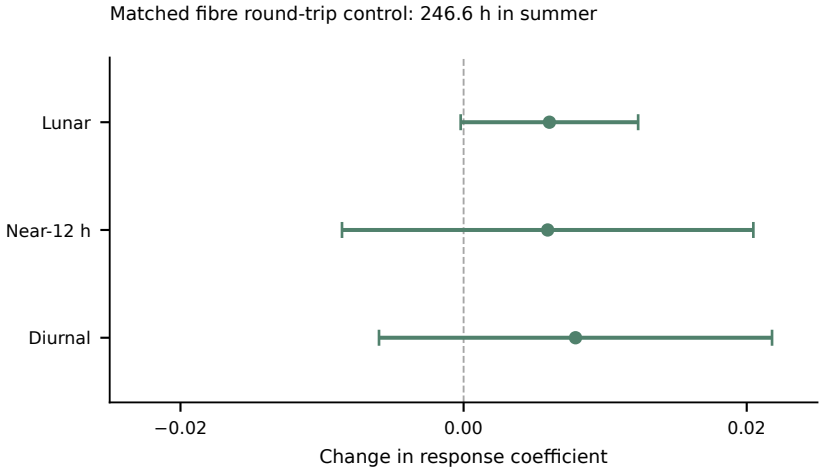
Extended Data Fig. 2 | Conditional semidiurnal scans. Equal-weight scans of the same 3,299 observations with 54 segment intercepts. **a,** M_2 neighbourhood before and after projection of a sine–cosine pair at S_2. **b,** S_2 neighbourhood before and after projection at M_2. Dotted lines show the astronomical target frequencies. These are descriptive projections on shared observations, not additional detection probabilities.



Extended Data Fig. 3 | Semidiurnal responses in eight individual observing periods. **a,** Lunar M_2+N_2 coefficients. **b,** Neighbouring S_2+K_2 coefficients. Each point uses only the indicated period, jointly estimates all three physical groups, and retains independent reference levels for its original segments. All eight periods are shown in calendar order, with 163, 203, 409, 447, 429, 601, 595 and 452 ten-minute means, respectively. Error bars show one statistical standard error, the larger of scaled least squares and a UTC-day-deletion jackknife computed within that period. The periods contain 3, 3, 4, 5, 5, 9, 8 and 6 observed UTC dates. Both axes are linear and show the full error ranges; their scales differ. Dashed lines mark the unit prediction. The plotted periods use disjoint observations and are fitted separately; the full-record estimate is obtained by the common physical fit.



Extended Data Fig. 4 | Same-data comparison of reference-level designs. a, Weighted segment-mean residuals after the eight-period joint fit, ranging from -6.75 to 12.00×10^{-18}. Segment constants make these means zero by construction. **b,** Lunar point estimates after period deletion, spanning 0.877–1.052 with eight period constants and 0.885–1.093 with 54 segment constants. The weighted residual sums of squares are 3,781.58 and 3,713.64; the latter has 46 additional intercept parameters. Unweighted residual root-mean-square values are 24.114 and 23.934×10^{-18}, and reduced residual sums of squares are 1.1501 and 1.1455. Adjacent standardized residual correlations are −0.127 and −0.148, using pairs within the same segment and separated by at most 900 s. All comparisons retain identical observations, templates, corrections and noise weights. These are residual diagnostics, not measurements of physical jumps or zero-transfer calibration.



Extended Data Fig. 5 | Simultaneous summer transmission control. Changes caused by adding the fibre round-trip monitor to the joint model, with paired UTC-day-jackknife standard errors. Baseline and augmented fits use identical clock/monitor members, labels and weights: 1,482 means containing 887,882 nominal one-second members, or 246.634 h in P6–P8, from 29 June to 25 August 2026. The lunar, neighbouring and daily shifts are respectively 0.00606 ± 0.00627, 0.00594 ± 0.01453 and 0.00791 ± 0.01388. The paired monitor constrains the measured transmission route on this summer overlap. The main tidal estimate uses the entire 545.9-h clock record. The prior transmission-architecture demonstration over 2,067 km is a separate experiment.[17]

Extended Data Table 1 | Astronomical constituent guide. Names specify forcing and frequency; one response coefficient multiplies each complete group with its internal ratios and phases fixed.

Symbol	Period_h	Meaning	Group
M2	12.421	Principal lunar semidiurnal	Lunar
N2	12.658	Lunar elliptic semidiurnal	Lunar
S2	12	Principal solar semidiurnal	Near-12 h
K2	11.967	Lunisolar semidiurnal	Near-12 h
K1	23.934	Lunisolar diurnal	Diurnal
O1	25.819	Principal lunar diurnal	Diurnal
P1	24.066	Principal solar diurnal	Diurnal
Q1	26.868	Lunar elliptic diurnal	Diurnal

603 **Extended Data Table 2 | Estimator and uncertainty comparison.** All directly replayed designs use the same 3,299 means, physical templates and
604 Allan-based weights. Errors are standard errors.

Design	Lunar	Near_12h	Diurnal	Status
S54	1.000 ± 0.132	0.925 ± 0.365	0.579 ± 0.538	Replayed
G29	0.984 ± 0.124	1.093 ± 0.298	0.759 ± 0.420	Replayed
P8	0.987 ± 0.123	1.129 ± 0.288	0.678 ± 0.359	Replayed

Group	Estimate	Scaled_LS	Day_JK	Period_JK	Reported_SE
Lunar	1.0004	0.132	0.1218	0.147	0.132
Near-12 h	0.9251	0.2993	0.3649	0.3531	0.3649
Diurnal	0.5792	0.3887	0.5381	0.7735	0.5381

Extended Data Table 3 | Conditional 95% interval coverage. Each scenario has 1,000 simulated realizations per design, with fixed observed sampling and weights. Values are percentages for the larger-of-LS-and-day-jackknife rule, using multiplier 1.96. They assess the specified scenarios, not unknown systematic effects.

scenario	design	Diurnal	Lunar	Near12h
50% OU 24 h, period	P8	97.9	98	97.3
50% OU 24 h, period	S54	96.4	95.1	95.4
50% OU 6 h, segment	P8	94.7	95.7	96
50% OU 6 h, segment	S54	95.8	95.6	96.3
Student t (df 5)	P8	96.8	96.3	96.2
Student t (df 5)	S54	96.2	96.1	96.8
Gaussian	P8	95.7	96.2	95.9
Gaussian	S54	95.5	97.4	96.7

608 **Extended Data Table 4 | Same-data 12-h sensitivity.** Each row retains the original 3,299 means, 54 segment levels, astronomical group templates
609 and Allan weights. Errors are the larger of scaled LS and day-deletion jackknife standard errors. Working intervals are estimate ±1.96 standard errors;
610 they are not calibrated for all correlated-noise or deterministic-disturbance models. A common 12-h sine–cosine pair has one phase across the record; the
611 period-wise design gives each of eight periods its own pair. These fits assess degeneracy, not the independently established magnitude of an instrumental
612 effect.

Model	Lunar coefficient	Lunar working interval	Neighbouring coefficient	Neighbouring working interval
Primary	1.000 ± 0.132	[0.742, 1.259]	0.925 ± 0.365	[0.210, 1.640]
Shared 12 h	0.994 ± 0.132	[0.735, 1.252]	2.931 ± 1.011	[0.950, 4.913]
12 h per period	0.751 ± 0.242	[0.277, 1.224]	-5.100 ± 15.008	[-34.515, 24.316]

Extended Data Table 5 | Full-template response and sensitivity. Each fit uses the same 3,299 ten-minute means, 54 segment constants and Allan-based weights. The scalar response multiplies the complete solid-Earth and ocean-loading waveform, including the minor-constituent remainder. Standard errors are statistical; the reported value is the larger of scaled least squares and UTC-day-deletion jackknife. The grouped-model sensitivities remain in Extended Data Table 4.

Model	Response	Scaled LS	Day jackknife	Reported SE
Whole template	0.9551	0.0973	0.1082	0.1082
With shared 24-h pair	1.0316	0.1028	0.098	0.1028
With period-specific 12-h pairs	0.7222	0.1791	0.2212	0.2212